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SEAT No. _____

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SARDAR PATEL UNIVERSITY
M Sc IT II Semester Examination
PS02CINT53: Artificial Intelligence

Date: 1/11/2023

Time: 10:00 a.m. to 1:00 p.m.

Total Marks: 70

Q1 Choose the most appropriate option for each question.

[8]

- 1 Which of the following is not a constituent of the AI definition?
 (A) Heuristic methods (B) Human thought process (C) Algorithmic (D) Non-algorithmic
- 2 _____ is a rule of thumb, that is practically applicable however, it does not offer a guarantee of a solution.
 (A) Rules (B) Heuristics (C) Facts (D) All of the above
- 3 Which of the following logic is the form of fuzzy logic?
 (A) Two-valued (B) Crisp set (C) Multi-valued (D) All of the above
- 4 In fuzzy logic, the AND operator represents the _____.
 (A) Minimum (B) Maximum (C) Complement (D) Both A and B
- 5 ANN requires _____ to learn.
 (A) Data (B) Model (C) Learning algorithm (D) All of these
- 6 ANN learns in _____ manner.
 (A) Parallel (B) Asynchronous (C) Distributed (D) All of these
- 7 GA is based on _____ principle.
 (A) Survival of the fittest (B) Heuristic (C) Logic (D) None of these
- 8 Order of the schema **001 in the genetic algorithm is _____.
 (A) 1 (B) 2 (C) 3 (D) None of these

Q2 Answer the following questions (Any Seven):

[14]

- 1 List any four characteristics of natural intelligence.
- 2 Name two strategies of inference engine working.
- 3 Define de-fuzzification.
- 4 List knowledge representation schemes.
- 5 List operations on fuzzy sets.
- 6 Write a rule for updating a node in the Hopfield network.
- 7 Give any two activation functions of an artificial neuron.
- 8 List any two stopping criteria of the genetic algorithm.
- 9 List two applications of the genetic algorithm.

Q3 Answer the following questions:

A List three types of AI application areas. Also, give examples from each category. Which area is the most difficult to implement? [6]

B List the categories of knowledge-based systems. Briefly explain any one category. [6]

OR

B Explain the knowledge acquisition process. [6]

Q4 Answer the following questions:

A Explain any two fuzzy membership functions. [6]

B What is a Fuzzy Logic Controller (FLC)? What are the steps involved in designing FLC? [6]

OR

B Explain fuzzy relations with an example. [6]

Q5 Answer the following questions:

A Draw a perceptron of your choice. Explain its learning process in detail. [6]

B Write a short note on multi-layer perceptron with an example with training data. [6]

OR

B Design and discuss a multi-layer perceptron for prediction of sales for the fast moving consumer goods. Take at least 4 to 5 input parameters. Also, provide a few training data sets. [6]

Q6 Answer the following questions:

A Write a short note on the genetic algorithm by giving a flowchart of the algorithm. [6]

B Explain how the genetic algorithm solves a function optimization problem by taking an example. [6]

OR

B Explain binary encoding schemes with different genetic operators. [6]

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